Sheet 7

1.

linear ODE $\quad f(t,u(t)) = \overset{\text{Matrix}}{A(t)}\,\overset{\text{Vector}}{u(t)} + \overset{\text{Vector}}{b(t)}$

quadratic ODE $\quad f(t,u(t)) = u(t)A(t)u(t)+b(t)$

$u(t) \in \mathbb{R}^2 \quad t \in \mathbb{R} \quad B \in \mathbb{R}^{2\times 4}$

$\frac{\partial u(t)}{\partial t}$	linear	non-linear
time invariant	$I\,u(t)$	$B(u(t) \otimes u(t))$
time variant	$(tI)\,u(t)$	$(tB)(u(t) \otimes u(t))$

$\otimes$ represents the Kronecker-Product.

e)

$$\frac{\partial u(t,x)}{\partial t} + \frac{\partial u(t,x)}{\partial x} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

3. a)

$$\frac{\partial u(t)}{\partial t} = f(t, u(t))$$

Neural ODE $\longrightarrow$ Classifier $\longrightarrow$ Loss

- final state $u(T)$ is given to a classifier (i.e. linear)
- classifier is trained via a loss (i.e. crossentropy)

3. b)